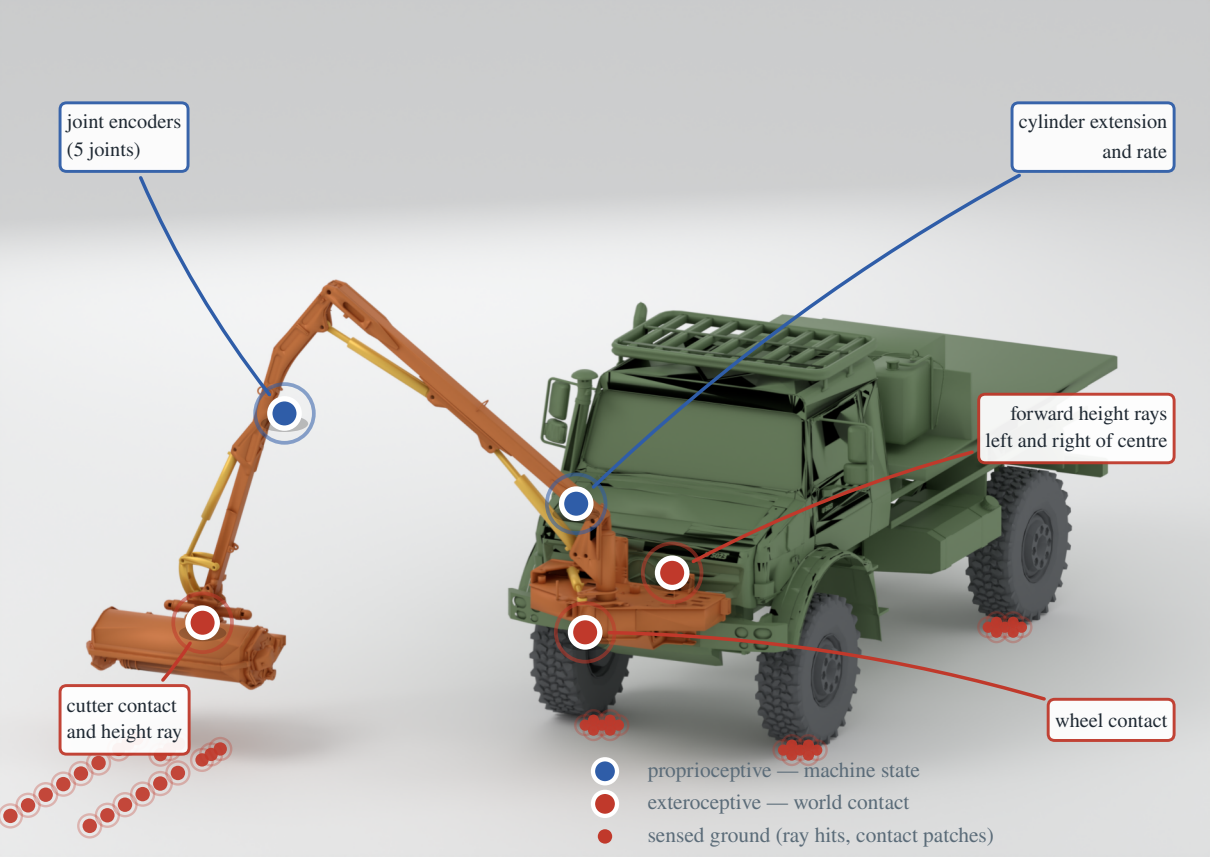

joint encoders
(5 joints)
cylinder extension
and rate
forward height rays
left and right of centre
cutter contact
and height ray
wheel contact
proprioceptive — machine state
exteroceptive — world contact
sensed ground (ray hits, contact patches)